R-PERFECT COMPLEXES STAY R'-PERFECT AFTER BASE CHANGE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The derived base change of a pseudo-coherent complex along a ring map is pseudo-coherent. invoked at proof lines 22030-22032. Gives the pseudo-coherence half of relative perfectness.
- The two derived tensor products K tensor_A A' and K tensor_R R' agree in D(R') for the base change diagram. invoked at proof lines 22033-22034. Identifies the object whose tor dimension is to be checked.
- The derived pullback of a complex of finite tor dimension has finite tor dimension. invoked at proof lines 22035-22037. Completes the finite tor dimension half.

Source reference. Original lemma 0650: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16345–16345. Cited at proof lines 22030-22031 for the pseudo-coherence of the base change; the conclusion is used inside the counted span.

Source reference. Original lemma 0661: sources/stacks-more-algebra/pinned-original-source.tex, source lines 14955–14955. Cited at proof lines 22033-22034 to identify K tensor_A A' with K tensor_R R' in D(R'); the conclusion is used inside the counted span.

Source reference. Original lemma 066L: sources/stacks-more-algebra/pinned-original-source.tex, source lines 17148–17148. Cited at proof lines 22035-22037 for the finite tor dimension after base change; the conclusion is used inside the counted span.

Lemma 0.2. *Let $R \to A$ be a flat ring map of finite presentation. Let $R \to R'$ be a ring map and set $A' = A \otimes_R R'$. If $K \in D(A)$ is R-perfect, then $K \otimes^{\mathbf{L}}_A A'$ is R'-perfect.*

Proof. By Lemma 0650 we see that $K \otimes^{\mathbf{L}}_A A'$ is pseudo-coherent. By Lemma 0661 we see that $K \otimes^{\mathbf{L}}_A A'$ is equal to $K \otimes^{\mathbf{L}}_R R'$ in $D(R')$. Then we can apply Lemma 066L to see that $K \otimes^{\mathbf{L}}_R R'$ in $D(R')$ has finite tor dimension. □

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